# Honda Civic e:HEV Dual-Motor Hybrid High Voltage Service Manual

Manufacturer: **Honda** | Model: **Civic** | Model Year: **2024** | Type: **emergency_guide**
Engine Architecture: **2.0L Direct Injection Atkinson-Cycle** | Transmission: **Electronic Continuously Variable Transmission (e-CVT)**
Official Source Repository: https://owners.honda.com/hybrid/civic-ehev-service-guide/

## 1. Technical Specifications & Capacities

| Parameter | Factory Specification | Inspection Interval |
|---|---|---|
| Engine Oil Type | 0W-20 / 5W-30 API SP/SN Full Synthetic | Every 5,000 km (Pakistan Severe Profile) |
| Engine Oil Sump Capacity | 3.7 - 4.2 Litres (with filter replacement) | Check weekly on level ground |
| Cooling System Coolant | Ethylene Glycol 50/50 Pre-mixed (Long Life) | Inspect level monthly, replace 80,000 km |
| Transmission Lubricant | Electronic Continuously Variable Transmission (e-CVT) | Inspect every 20,000 km / Replace 40,000 km |
| Brake Fluid Specification | DOT 3 / DOT 4 Polyethylene Glycol | Inspect level monthly, flush every 2 years |
| Cold Tyre Pressure | 32 PSI (Front) / 32 PSI (Rear) | Inspect weekly when tyres are cold |

## 2. Critical Safety Instructions & Warning Indicators

**WARNING: ENGINE OVERHEATING EMERGENCIES**
**If the engine coolant temperature gauge enters the RED zone, or steam escapes from under the hood: Immediately pull over to a safe area away from moving traffic and shift transmission into Park/Neutral. Turn off air conditioning immediately. DO NOT open the radiator cap or pressurized coolant reservoir while hot. Scalding pressurized coolant will cause catastrophic skin burns. Wait at least 30 minutes until the engine is completely cold before inspecting the coolant level or checking for burst radiator hoses.**

**CAUTION: ENGINE OIL PRESSURE WARNING LIGHT**
**The oil lamp indicator signals a loss of oil pressure, NOT oil level. If illuminated while driving, stop the vehicle immediately and turn off the engine. Continued operation under zero or low oil pressure will seize bearings and permanently destroy the engine block within 30 seconds. Verify oil dipstick level and check for catastrophic oil filter or sump puncture.**

**DANGER: BRAKE SYSTEM WEAR & BRAKE WARNING LIGHT**
**Brake pads include built-in acoustic wear indicators. When pad thickness drops below 1.0 mm, a metallic screeching sound occurs during braking. If the red brake warning light remains lit with the parking brake fully released, a hydraulic leak or dangerously low fluid level is present. Do not drive the vehicle. Arrange emergency towing.**

## 3. Vehicle Subsystem Architecture & Diagnostic Procedures

- **Subsystem Guideline:** High-voltage disconnect procedure in rear cargo area
- **Subsystem Guideline:** Power Control Unit (PCU) coolant maintenance
- **Subsystem Guideline:** Emergency response for collision damage with damaged orange harnesses

## 4. Diagnostic Trouble Code (DTC) Matrix Reference

| OBD-II Code | Fault Description | Primary Mechanical Root Causes |
|---|---|---|
| P0300 | Random / Multiple Cylinder Misfire Detected | Failing ignition coils, worn spark plugs, low fuel pressure, vacuum leak |
| P0171 | System Too Lean (Bank 1) | Dirty MAF sensor, vacuum intake leak, clogged fuel injectors, weak fuel pump |
| P0420 | Catalytic Converter System Efficiency Below Threshold | Exhaust leak ahead of converter, failing downstream O2 sensor, carbon fouling |
| P0117 | Engine Coolant Temperature Circuit Low Input | Stuck closed thermostat, failed ECT sensor, short in wiring harness |
| P0700 | Transmission Control System Malfunction | Degraded CVT fluid, valve body solenoid failure, overheated fluid sensor |

**Note on Verification:** Workshop repair estimates that recommend major replacements (e.g., complete transmission or catalytic converter replacement) must be verified against actual scanner live sensor data and basic component checks (e.g. spark plugs, fluid age, wiring continuity) before authorization.